Pairwise Independence of Representation, Classification, and Composition in Finite Extensional Magmas

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Abstract

Nontrivial combinatory algebras with S and K must be infinite. Associativity is incompatible with combining a classifier and a retraction pair in a finite extensional magma. These obstructions exclude several standard settings from the finite extensional framework studied here, most notably nontrivial finite S+K-style combinatory algebras and associative structures (semigroups, monoids, groups, rings) carrying both a classifier and a retraction pair. What algebraic structure exists in the remaining landscape: finite, non-associative, total?

We identify three properties of finite extensional 2-pointed magmas: self-representation (R), the classifier dichotomy (D), and the Internal Composition Property (H). We prove they are pairwise independent. Six Lean-verified finite counterexamples at sizes 5 through 10 establish all six non-implications. The minimum coexistence witness has $N=5$, which is optimal: ICP requires 3 pairwise distinct core elements, so $N \geq 5$. The three-category decomposition induced by D is an isomorphism invariant, and the ICP is logically equivalent to the standard Compose+Inert axioms. All results are formalized in Lean 4 with zero `sorry`.

Keywords

finite extensional magma, pairwise independence, classifier dichotomy, internal composition, Lean 4, machine-checked proofs

1 Introduction

Nontrivial combinatory algebras with **s** and **k** are necessarily infinite (Theorem 3.7). Associativity is incompatible with combining a classifier and a retraction pair in a finite extensional magma (Theorem 3.6). These two obstructions carve out a specific algebraic landscape: finite, non-associative, total. They rule out the most immediate finite instantiations of PCA [13, 14], TCA [15], and Turing-object [17] universality (by cardinality), and they exclude associative settings such as semigroups, monoids, groups, and rings from supporting the R+D pattern considered here. What algebraic structure exists in the remaining landscape?

We work in the setting of finite extensional 2-pointed magmas (a finite set with a binary operation, exactly two left-absorbers, and extensionality; the framework follows universal algebra [10]). We identify three algebraic properties: self-representation (R), the existence of a section-retraction pair on the non-absorber elements (the *core*); the classifier dichotomy (D), a clean partition of core elements into those whose action on core lands entirely in the absorbers and those whose action avoids them entirely; and the Internal Composition Property (H), the existence of a non-trivial factorization in the left-regular representation restricted to core.

Our main result is that R, D, and H are pairwise independent: no one implies any other. Six Lean-verified finite counterexamples of sizes $N=5$, 6, 8, and 10 establish all six non-implications; coexistence witnesses at $N=5$ and $N=6$ show all three can hold simultaneously. The $N=5$ bound is optimal: ICP requires 3 pairwise distinct core elements, forcing $|S| \geq 5$. Each counterexample is a concrete Cayley table found by SAT search and independently verified in Lean 4 [12], zero `sorry`.

Informally, R, D, and H correspond to the three components of self-simulation: encoding, classification, and internal composition. Reflective systems in practice combine all three: Smith's 3-Lisp [1], Friedman and Wand's reification [2], Amin and Rompf's collapsed towers [4], and Brown and Palsberg's typed self-interpreters [6] all bundle representation, classification, and composition without separating them. In the finite algebraic setting, we show that this bundling is not structurally forced; whether the independence extends to λ-calculus or typed settings is an open question (Section 9).

Beyond independence, we prove several additional structural results. The no-associativity theorem (Theorem 3.6) rules out the classical algebraic hierarchy: no extensional 2-pointed magma carrying both a classifier and a retraction pair has a right identity or is associative. The three-category decomposition induced by D is an algebraic invariant (Theorem 4.2), and the ICP is equivalent to the standard Compose+Inert axioms (Theorem 5.1).

Contributions.

(1) Universal algebraic obstructions that motivate the setting: K-infinity rules out finite combinatory algebras; no-associativity rules out the classical algebraic hierarchy (Section 3).
(2) An independence theorem: R, D, and H are pairwise independent, with Lean-verified counterexamples in all six directions at $N = 5$ through $N = 10$ (Section 6), and coexistence witnesses at $N=5$ and $N=6$ (Section 7).
(3) An optimal bound: $N = 5$ is the minimum carrier size for R+D+H coexistence. ICP is impossible at $N=4$ by pigeonhole (Theorem 7.1).
(4) Decomposition invariance: the three-category partition is preserved by isomorphisms (Section 4).
(5) A characterization of H confirming the ICP is not ad hoc: it is logically equivalent to Compose+Inert under variable renaming, with necessity of the non-degeneracy conditions demonstrated by Lean-verified witnesses (Section 5).

2 Definitions

Throughout, $(S, \cdot)$ is a finite magma on $\mathrm{Fin}(n) = \{0, 1, \ldots, n-1\}$ with binary operation $\cdot : \mathrm{Fin}(n) \times \mathrm{Fin}(n) \to \mathrm{Fin}(n)$.

Preprint version. The Lean proofs, SAT search scripts, and manuscript text were developed with assistance from Claude (Anthropic). All results have been independently verified: Lean proofs compile with `lake build` (zero `sorry`), and SAT counterexamples are frozen in the repository for reproducibility. The author takes full responsibility for all contents.

Definition 2.1 (Extensional 2-Pointed Magma). An *extensional 2-pointed magma* is a tuple $(S, \cdot, z_1, z_2)$ where:

(1) $z_1, z_2 \in S$ are distinct *left-absorbers*: $z_i \cdot x = z_i$ for all x.
(2) No other element is a left-absorber.
(3) *Extensionality*: if $a \cdot x = b \cdot x$ for all x, then $a = b$.

We write $\mathrm{core}(S) := S \setminus \{z_1, z_2\}$ for the non-absorber elements.

Definition 2.2 (Faithful Retract Magma). A *Faithful Retract Magma* (FRM) extends an extensional 2-pointed magma with a *retraction pair* $s, r \in \mathrm{core}(S)$ satisfying $r \cdot (s \cdot x) = x$ for all $x \in \mathrm{core}(S)$, with $r \cdot z_1 = z_1$ (anchoring). The remaining absorber outputs of s and r are unconstrained.

Remark 1 (Retraction pair convention). Throughout this paper, the retraction pair (s, r) satisfies mutual inverse on core: $r \cdot (s \cdot x) = x$ and $s \cdot (r \cdot x) = x$ for all $x \in \mathrm{core}(S)$. This matches the Lean formalization. Most theorems (9 of 11) require only the one-sided condition $r \cdot (s \cdot x) = x$; two results, retraction pair placement (both $s, r \in N$; Theorem 3.3) and the tight bound $|S| \geq 5$ when $s \neq r$ (Theorem 3.4), use the full mutual inverse. We adopt the stronger convention for uniformity and note the weaker hypotheses where relevant.

The retraction pair provides a section/retraction structure on core: s acts as a section and r as a retraction. In finite witnesses this can be used to organize elements into coding schemes, but we do not assume a canonical global encoding.

Definition 2.3 (Classifier Dichotomy). An extensional 2-pointed magma satisfies the *classifier dichotomy* when:

(1) A *classifier* $\tau \in \mathrm{core}(S)$ exists with $\tau \cdot x \in \{z_1, z_2\}$ for all x.
(2) For every $y \in \mathrm{core}(S)$, either
   - $y \cdot x \in \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$ (y is a *classifier*), or
   - $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$ (y is a *non-classifier*).
(3) At least one non-classifier exists (excluding the degenerate case where the "dichotomy" has only one inhabited class).

The dichotomy creates a three-category decomposition $S = Z \sqcup C \sqcup N$ where $Z = \{z_1, z_2\}$ (absorbers), C (classifiers), and N (non-classifiers). No element can be partially classifying. We call elements in $\{z_1, z_2\}$ *absorber-valued* (abbreviated *boolean-valued* when the absorbers play the role of truth values).

Definition 2.4 (Internal Composition Property). A magma $(S, \cdot, z_1, z_2)$ satisfies the *Internal Composition Property* (ICP) when:

$$\exists\, a, b, c \in \mathrm{core}(S), \text{ pairwise distinct, such that:}$$

(1) *Core-preservation*: $b \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$.
(2) *Factorization*: $a \cdot x = c \cdot (b \cdot x)$ for all $x \in \mathrm{core}(S)$.
(3) *Non-triviality*: $|\{a \cdot x : x \in \mathrm{core}(S)\}| \geq 2$.

ICP says: the left-regular representation admits a non-trivial factorization on core. One core action factors through two others, one of them core-preserving: a single fragment of internal composition. In Section 5 we prove that ICP is equivalent to the standard operational axioms Compose+Inert (Theorem 5.1).

Definition 2.5 (Capability R). An extensional 2-pointed magma has capability R (self-representation) if it admits a retraction pair.

Definition 2.6 (Capability D). An extensional 2-pointed magma has capability D (self-description) if it satisfies the classifier dichotomy (Definition 2.3: existence of a classifier, the global dichotomy, and non-degeneracy).

Definition 2.7 (Capability H). An extensional 2-pointed magma has capability H (self-execution) if it satisfies the ICP.

All three capabilities are properties of the same base structure (Definition 2.1). None is defined as an extension of any other, making the independence theorem (Section 6) a statement about six non-implications, all proved by Lean-verified counterexamples. The labels "self-representation," "self-description," and "self-execution" are heuristic names for the formal properties R, D, and H; no claim is made that these structures by themselves implement a full internal evaluator.

3 Universal Theorems

The following theorems are proved by pure algebraic reasoning (no `decide`) and formalized in `CatKripkeWallMinimal.lean` and `NoCommutativity.lean`. We state the hypotheses explicitly: some require only D, while others use both R and D.

3.1 Theorems from D alone

Theorem 3.1 (Three-Category Decomposition). *In any extensional 2-pointed magma satisfying D, every element is a zero, a classifier, or a non-classifier. The classes are pairwise disjoint: in particular, $C \cap N = \emptyset$ (no element is partially classifying).*

Theorem 3.2 (Asymmetry). *No extensional magma with two distinct left-absorbers is commutative.*

3.2 Theorems from R+D

Theorem 3.3 (Retraction Pair Placement). *In any extensional 2-pointed magma satisfying both R and D, any witness r to R is a non-classifier: $r \in N$. (Under the stronger mutual-inverse condition in the Lean formalization, $s \in N$ also holds.)*

Theorem 3.4 (Cardinality Bounds). *$|S| \geq 4$ for any extensional 2-pointed magma satisfying both R and D. (The bound $|S| \geq 4$ is nearly immediate from the definitions, since the four roles (two absorbers, one classifier, one non-classifier) must be distinct; the Lean-verified proof confirms that no subtle interaction reduces it. The bound is tight: a 4-element witness with $s = r$ is exhibited in Appendix A.) Under the stronger mutual-inverse condition with $s \neq r$, $|S| \geq 5$ (also tight); this is the nontrivial bound, as it shows the retraction pair consumes two distinct core slots.*

Theorem 3.5 (No Right Identity). *No extensional 2-pointed magma satisfying both R and D has a right identity element.*

Theorem 3.6 (No Associativity). *No extensional 2-pointed magma carrying both a classifier and a retraction pair is associative. In particular, no extensional 2-pointed magma satisfying both R and D is a semigroup, monoid, group, or ring: the classifier dichotomy with a retraction pair is incompatible with the entire classical algebraic hierarchy.*

Proof. Assume $(\tau \cdot a) \cdot b = \tau \cdot (a \cdot b)$ for all a, b. We derive a contradiction in four steps. The proof uses only extensionality, the

existence of a classifier τ, and a retraction pair (s, r). Neither the global dichotomy nor the non-classifier existence axiom is needed. Lean file: `CatKripkeWallMinimal.lean`.

Step 1: τ absorbs on the right. For any a, b:

$$\tau \cdot (a \cdot b) = (\tau \cdot a) \cdot b = z_i \cdot b = z_i = \tau \cdot a$$

where the second equality uses $\tau \cdot a \in \{z_1, z_2\}$ (classifier), and the third uses left-absorption. So $\tau \cdot (a \cdot b) = \tau \cdot a$ for all a, b.

Step 2: τ is constant on core. For any core x: the retraction pair gives $x = r \cdot (s \cdot x)$. Applying Step 1 with $a = r$, $b = s \cdot x$:

$$\tau \cdot x = \tau \cdot (r \cdot (s \cdot x)) = \tau \cdot r.$$

Set $v = \tau \cdot r \in \{z_1, z_2\}$. Then $\tau \cdot x = v$ for all core x.

Step 3: τ is constant on absorbers. Since τ is a classifier, it hits both z_1 and z_2; otherwise, if $\tau \cdot x = z_1$ for all x, then τ and z_1 have identical rows, so $\tau = z_1$ by extensionality, contradicting $\tau \neq z_1$ (and symmetrically for z_2). Let b_1, b_2 witness $\tau \cdot b_1 = z_1$ and $\tau \cdot b_2 = z_2$. Applying Step 1 with $a = \tau$:

$$\tau \cdot z_1 = \tau \cdot (\tau \cdot b_1) = \tau \cdot \tau = v, \qquad \tau \cdot z_2 = \tau \cdot (\tau \cdot b_2) = \tau \cdot \tau = v.$$

The middle equalities use Step 1; $\tau \cdot \tau = v$ because τ is core (Step 2).

Step 4: Contradiction. Steps 2 and 3 give $\tau \cdot x = v$ for all $x \in S$. Since $v \in \{z_1, z_2\}$, the classifier τ has the same row as absorber v. By extensionality, $\tau = v \in \{z_1, z_2\}$, contradicting $\tau \neq z_1, z_2$. □

Remark 2. The retraction pair is essential in Step 2: the identity $\tau \cdot x = \tau \cdot (r \cdot (s \cdot x)) = \tau \cdot r$ uses $r \cdot (s \cdot x) = x$ to show τ is constant on core. Without a retraction pair, the proof breaks, and the theorem does not extend to "no extensional 2-pointed magma with a classifier is associative."

3.3 The K-infinity obstruction

The following classical result explains why the PCA/TCA framework cannot study finite algebras, and why our setting requires a different combinator basis.

Theorem 3.7 (K-infinity). *If $(A, \cdot, \mathbf{k}, \mathbf{s})$ is a combinatory algebra (total application satisfying $\mathbf{k} \cdot a \cdot b = a$ and the $\mathbf{s}$ equations) with $|A| \geq 2$, then A is infinite.*

Proof. The map $\varphi(x) = \mathbf{k} \cdot x$ is injective: $\mathbf{k} \cdot a = \mathbf{k} \cdot b$ implies $\mathbf{k} \cdot a \cdot c = \mathbf{k} \cdot b \cdot c$ for all c, hence $a = b$. We claim $\mathbf{k} \notin \mathrm{Im}(\varphi)$. Suppose $\mathbf{k} \cdot x = \mathbf{k}$ for some x. Then for all y, $\mathbf{k} \cdot x \cdot y = \mathbf{k} \cdot y$, so $x = \mathbf{k} \cdot y$ for all y. By injectivity, all y are equal, contradicting $|A| \geq 2$. Thus $\varphi : A \to A$ is an injection missing $\mathbf{k}$, which is impossible for finite A. □

Remark 3. The proof uses only the $\mathbf{k}$ equation ($\mathbf{k} \cdot a \cdot b = a$) and totality; $\mathbf{s}$ is irrelevant. Our extensional 2-pointed magmas avoid this obstruction: instead of a projector $\mathbf{k}$ with $\mathbf{k} \cdot a \cdot b = a$, they have left-absorbers z with $z \cdot x = z$ (the dual pattern, constant on columns rather than rows) and a retraction pair (s, r) with $r \cdot (s \cdot x) = x$ on core. Neither structure provides a non-surjective injection on the full ground set.

4 Decomposition Invariance

Definition 4.1 (Absorber-Preserving Isomorphism). An *absorber-preserving isomorphism* $\varphi : M_1 \to M_2$ between extensional 2-pointed magmas is a bijection $\varphi : \mathrm{Fin}(n) \to \mathrm{Fin}(n)$ satisfying:

- $\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b)$ for all a, b
- $\varphi(z_1^{(1)}) = z_1^{(2)}$ and $\varphi(z_2^{(1)}) = z_2^{(2)}$

Theorem 4.2 (Functoriality). *Any absorber-preserving isomorphism between extensional 2-pointed magmas satisfying D preserves the three-category decomposition: φ maps zeros to zeros, classifiers to classifiers, and non-classifiers to non-classifiers.*

Proof sketch. *Zeros*: if a is a left-absorber, then for all x in the image of φ, $\varphi(a) \cdot_2 \varphi(x) = \varphi(a \cdot_1 x) = \varphi(a)$. By surjectivity, $\varphi(a)$ is a left-absorber.

Classifiers: if $y \cdot_1 x \in \{z_1, z_2\}$ for all core x, then $\varphi(y) \cdot_2 \varphi(x) = \varphi(y \cdot_1 x) \in \{z_1', z_2'\}$ for all core $\varphi(x)$. By surjectivity and injectivity (to exclude absorber preimages), $\varphi(y)$ is a classifier.

Non-classifiers: symmetric argument using injectivity to transport $\notin \{z_1, z_2\}$.

Algebraic proof (no `decide`); Lean file: `Functoriality.lean`. □

5 ICP Characterization

An algebra with R can encode, decode, and look up any entry in the Cayley table via the retraction pair (s, r). But this requires an *external* machine to supply control flow: dispatch, composition, recursion. H internalizes a non-trivial composition pattern that would otherwise be supplied externally: dedicated elements whose left-multiplication implements composition ($\eta = \rho \circ g$) and storage (g). The ICP captures exactly this: one core action factors through two others, meaning the algebra contains a fragment of its own control flow as an element.

The standard formulation uses two operational axioms:

- *Compose*: $\eta \cdot x = \rho \cdot (g \cdot x)$ for all $x \in \mathrm{core}(S)$
- *Inert*: $g \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$

with η, g, ρ pairwise distinct non-absorbers and η non-trivial (≥ 2 distinct values on core).

Theorem 5.1 (ICP Equivalence). *For any finite magma on a 2-pointed set $(S, \cdot, z_1, z_2)$ of any size n:*

$$\textit{ICP} \iff \textit{Compose} + \textit{Inert (with non-trivial composite).}$$

Proof. The bijection between witness triples is $(\eta, g, \rho) = (a, b, c)$: the composed element a is η, the core-preserving element b is g, and the outer element c is ρ. Under this identification the factorization $a \cdot x = c \cdot (b \cdot x)$ becomes $\eta \cdot x = \rho \cdot (g \cdot x)$ (Compose), core-preservation of b becomes Inert, and the remaining conditions (pairwise distinct, non-absorber) transfer directly.

The pairwise-distinctness and non-triviality conditions are not cosmetic; they block two degeneracies, and both are proved necessary by Lean-verified witnesses (`ICP.lean`). Without pairwise-distinctness, the retraction pair (s, r) with $s = r$ trivially satisfies factorization via $s \cdot x = s \cdot (s \cdot x)$, giving every FRM a spurious ICP; the minimal R+D model at $N = 4$ (Cayley table in Appendix A; Lean name `kripke4`) separates the weakened definition from the full one (theorem `icp_distinct_necessary`). Without non-triviality, any constant-on-core element a (with $a \cdot x = v$ for all core x) trivially factors through any core-preserving b by taking c with $c \cdot y = v$; a custom 6-element FRM separates the weakened definition from the full one (theorem `icp_nontrivial_necessary`, `ICP.lean`).

The proof is pure equational logic, with no `decide`, no case analysis on n. It holds for every finite magma on a 2-pointed set,

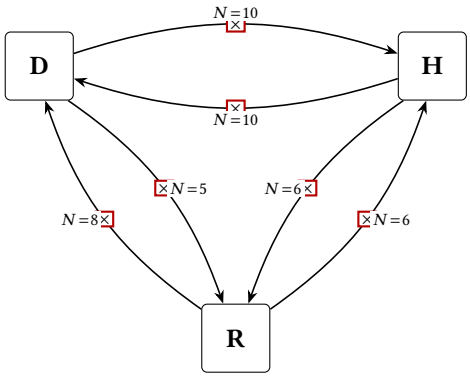

Figure 1: Full independence structure. All six pairwise non-implications are proved by Lean-verified finite counterexamples at the indicated sizes. No capability implies any other.

of any size. Lean file: `ICP.lean`, theorem `icp_iff_composeInert`. □

This equivalence gives H a single-concept definition: the left-regular representation contains a non-trivially composed element. The operational axioms (Compose, Inert) and the ICP are the same property under variable renaming. Additionally, ICP is invariant under FRM isomorphisms (algebraic proof, no `decide`; `ICP.lean`, theorem `FRMIso.preserves_icp`).

Enrichments (not used in this paper). Two additional axioms connect the capabilities and are used in the accompanying artifact: *Branch* ($\rho \cdot x = f \cdot x$ when $\tau \cdot x = z_1$, else $\rho \cdot x = g \cdot x$, for distinguished $f, \rho \in \mathrm{core}(S)$), which bridges D and H by making the classifier's judgment control evaluation dispatch; and a *fixed-point axiom* ($Y \cdot \rho = \rho \cdot (Y \cdot \rho)$ with $Y \cdot \rho \in \mathrm{core}(S)$), which extends H across the decidability boundary. Neither is required for the independence, invariance, or obstruction results of this paper.

6 Independence

Theorem 6.1 (Full Independence). *No capability implies any other. Specifically, all six pairwise non-implications hold:*

(1) *$R \not\Rightarrow D$: An extensional 2-pointed magma of size 8 satisfying R contains a classifier element but fails the global dichotomy condition.*
(2) *$R \not\Rightarrow H$: An extensional 2-pointed magma of size 6 satisfying R has 4 core elements (enough for ICP to be non-vacuous) but no triple satisfies the ICP.*
(3) *$D \not\Rightarrow H$: An extensional 2-pointed magma of size 10 satisfying both R and D has no triple satisfying the ICP.*
(4) *$H \not\Rightarrow D$: An extensional 2-pointed magma of size 10 satisfies ICP but violates the classifier dichotomy.*
(5) *$D \not\Rightarrow R$: An extensional 2-pointed magma of size 5 satisfies the classifier dichotomy but admits no retraction pair.*
(6) *$H \not\Rightarrow R$: An extensional 2-pointed magma of size 6 satisfies ICP but admits no retraction pair.*

Proof. Each direction is proved by exhibiting a concrete Cayley table and verifying the claimed properties by `native_decide` (or `decide` for $N \leq 8$). The tables are:

$R \not\Rightarrow D$ ($N = 8$, the Countermodel). An 8×8 Cayley table (Appendix A) defining an FRM with $s = 2$, $r = 3$. The retraction pair works: $r \cdot (s \cdot x) = x$ for all core $x \in \{2, \ldots, 7\}$ (e.g., $3 \cdot (2 \cdot 5) = 3 \cdot 6 = 5$). Element 4 is a classifier: its row on core is $(0, 0, 0, 0, 1, 0) \subseteq \{z_1, z_2\}$. But element 5's row on core is $(6, 2, 1, 1, 1, 1)$: it maps core inputs 2 and 3 to non-absorber values (6 and 2), while mapping inputs 4–7 to absorber value 1. This mixed behavior violates the classifier dichotomy: element 5 is neither entirely absorber-valued nor entirely non-absorber-valued on core. Self-representation holds; self-description fails. Lean file: `Countermodel.lean`.

$R \not\Rightarrow H$ ($N = 6$, structural). A 6×6 Cayley table (Appendix A) defining an extensional 2-pointed magma with a retraction pair ($s = r = 2$, an involution on core) but no ICP. The core has 4 elements, more than the 3 pairwise distinct witnesses ICP requires, yet no triple satisfies the ICP. Lean proves $\neg$ `HasICP` by exhaustive search via `decide`. Lean file: `E2PM.lean`, theorem `s_not_implies_icp_structural`.

$D \not\Rightarrow H$ ($N = 10$). A 10×10 Cayley table satisfying D (and also R). Lean proves that no triple (a, b, c) of pairwise distinct non-absorbers satisfies the ICP, by exhaustive search via `native_decide`. Lean files: `Countermodels10.lean`, `ICP.lean` (theorem `dNotH_no_icp`).

$H \not\Rightarrow D$ ($N = 10$). A 10×10 Cayley table defining an extensional 2-pointed magma with a retraction pair, satisfying ICP (witnessed by $a = 8, b = 6, c = 7$), yet violating the dichotomy: element 5 has both absorber-valued and non-absorber-valued outputs on core. Lean file: `Countermodels10.lean`.

$D \not\Rightarrow R$ ($N = 5$). A 5×5 Cayley table defining an extensional 2-pointed magma with the classifier dichotomy (classifier at index 2, non-classifiers at 3 and 4), but admitting no retraction pair. Lean proves $\neg$ `HasRetractPair` by exhaustive search over all candidate pairs via `decide`. Lean file: `E2PM.lean`, theorem `d_not_implies_s`.

$H \not\Rightarrow R$ ($N = 6$). A 6×6 Cayley table defining an extensional 2-pointed magma satisfying ICP (witnessed by $a = 2, b = 4, c = 5$), but admitting no retraction pair. Lean file: `E2PM.lean`, theorem `h_not_implies_s`.

All tables are generated by Z3 SAT solver and independently verified. The Cayley tables are frozen in the repository for reproducibility. □

Methodology. Each counterexample was found by encoding the relevant capability axioms (and the negation of the target capability) as a SAT problem over an $n \times n$ Cayley table, then solving with Z3. The search space is n^{n^2} (all possible binary operations on $\mathrm{Fin}(n)$); the axioms reduce this to a manageable constraint system. The sizes $N = 8$ and $N = 10$ were discovered by incrementing n from the minimum until SAT returned a model. Once found, each table was frozen in `counterexamples.json` and independently verified in Lean. The counterexample sizes are *not* claimed to be universally minimal: they are minimal among the role assignments tested (SAT returned UNSAT at all smaller sizes with the same role assignments), but different role assignments or overlapping encodings could in principle yield smaller witnesses. We state no lower bounds on counterexample size.

Interpretation. The independence result shows that R, D, and H are genuinely three things, not one. An algebra can encode and decode without classifying (R without D). An algebra can classify without encoding (D without R). An algebra can internally compose without classifying or encoding (H without R or D). No capability algebraically forces any other. In particular, the $H \nRightarrow D$ direction shows that an algebra can internalize a non-trivial composition pattern without satisfying the classifier dichotomy: internal composition does not force organizational separation.

7 Coexistence

Theorem 7.1 (No ICP at $N=4$). *No extensional 2-pointed magma of size 4 satisfies the ICP. The core $\{2,3\}$ has only 2 elements, but ICP requires 3 pairwise distinct non-absorbers. Lean:* `Witness5.lean`, `no_icp_at_4`.

Theorem 7.2 (Coexistence Witness). *The three capabilities R, D, and H are simultaneously satisfiable. A Lean-verified witness exists at $N=5$, which is optimal by Theorem 7.1.*

Proof. A concrete 5×5 Cayley table is exhibited with $s=r=2$ (identity on core), classifiers $\tau \in \{3,4\}$, and ICP witnessed by $a=3, b=2, c=4$. The retraction pair collapses to a single element ($s=r$) which doubles as the ICP's core-preserving map; since b is the identity, the factorization $a \cdot x = c \cdot (b \cdot x)$ reduces to $a \cdot x = c \cdot x$ on core, yielding two classifiers that agree on core but distinguished by their absorber columns. The lower bound is pigeonhole: at $N=4$, the core $\{2,3\}$ has only 2 elements, but ICP requires 3 pairwise distinct non-absorbers (`no_icp_at_4`). Lean file: `Witness5.lean`, theorem `sdh_witness_5`. A 6×6 witness with $s \neq r$ is also verified (`Witness6.lean`). □

Proposition 7.3 (Separated-Role Witness). *There is a witness at $N=10$ in which the principal R+D+H roles (2 absorbers, 2 retraction pair, 1 classifier, 3 ICP witnesses) occupy 8 distinct elements, with 2 additional elements serving the enrichment axioms (Branch and Y, defined in Section 5).*

Proof. A concrete 10×10 Cayley table (Appendix A) satisfies all three capabilities with 10 distinct role assignments. Additionally, the enrichments Branch and Y hold. Lean file: `Witness10.lean`. □

Minimum cardinality. $N=5$ is the optimal minimum for R+D+H coexistence (Lean-verified; `Witness5.lean`). The witness uses $s = r$ (identity on core), two classifiers, and one non-classifier: the retraction pair member doubles as the ICP's core-preserving element. At $N=4$ the core has only 2 elements but ICP requires 3 pairwise distinct non-absorbers, so $N=4$ is impossible (`no_icp_at_4`).

The $N = 5$ witness reveals how thin the three properties can become simultaneously. The retraction pair is the identity on core ($s = r$, no actual encoding work), and the ICP factorization $a \cdot x = c \cdot (b \cdot x)$ collapses to $a \cdot x = c \cdot x$ (row equality on core between two classifiers differing only on absorber columns). The three capabilities are formally present but barely interacting. The $N = 6$ witness with $s \neq r$ (`Witness6.lean`) is the more informative example: the retraction pair does genuine encoding, the ICP factorization is non-degenerate, and the roles do not trivially overlap. We retain the general definitions (allowing $s = r$) since the algebraic results (independence, injectivity, no-associativity) hold regardless.

8 Related Work

Partial and total combinatory algebras. PCAs [13, 14] and TCAs [15] provide the standard algebraic framework for realizability and computability. PCA theory has a rich structural theory: morphisms between PCAs, the lattice of PCA quotients, and totalizability [16] all involve detailed analysis of the application operation. PCA theory does study which elements realize which functions. However, this analysis concerns representability and inter-PCA relationships, not the *partition* of elements into behavioral types by their row structure. This question requires a finite total operation table to formulate; it does not arise in the PCA/TCA setting, where application is partial or the carrier is infinite. Bethke and Klop [16] proved that some PCAs cannot be extended to total ones, a structural result about the relationship between partiality and totality. The K-infinity theorem (our Theorem 3.7) shows that nontrivial S+K algebras must be infinite, explaining why finite algebras fall outside this framework and why our setting requires a different combinator basis.

Turing categories. Cockett and Hofstra [17] define a Turing category as a cartesian restriction category with a Turing object A and a Turing morphism $\bullet : A \times A \to A$ that universally simulates all morphisms in the category. The Structure Theorem establishes that PCAs and Turing categories are two views of the same concept. Turing objects have cardinality "one or infinite" [17], ruling out nontrivial finite Turing objects. The Turing morphism is described as "the heart of the whole structure" but its algebraic properties (row structure, element classification, associativity constraints) are not studied. Where Turing categories distribute computational roles across distinct objects (programs, data, evaluators), our finite algebras develop the same roles internally as emergent behavioral regions within a single carrier set: absorbers encode failure, classifiers provide absorber-valued judgment, and the retraction pair provides encoding.

Reflective towers. Smith's 3-Lisp [1] introduced the reflective tower, an infinite chain of meta-interpreters. Subsequent work [2, 3] explored reification, intensions, and live meta-modification in reflective towers. Amin and Rompf [4] showed how to compile through the tower via stage polymorphism. All of these systems combine self-representation, self-description, and self-execution without separating them. Our contribution is to show that in finite extensional 2-pointed magmas, the three capabilities are separable: no capability implies any other.

Self-interpretation. Mogensen [5] constructs self-interpreters for the λ-calculus. Brown and Palsberg [6] show that self-interpretation is possible in strongly-normalizing typed languages, contradicting earlier impossibility results. These works study self-interpretation (our R) but do not address the independence of R from D or H.

Independence in finite algebra. Independence and non-implication results between equational properties have a long history in universal algebra, from Birkhoff's characterization of equational classes [10] to Mal'cev conditions characterizing implications between properties of varieties. Our results operate at a different level: we separate *operational* properties (retraction pairs, classifier partitions,

factorizations in the left-regular representation) rather than equational identities, and our base class, the finite extensional 2-pointed magmas, is not a variety in the Birkhoff sense (extensionality is a quasi-identity, not an equation). To our knowledge, the pairwise independence of retraction pairs, classifier dichotomies, and factorization properties in finite magmas has not been previously established. The SAT-find-then-Lean-verify methodology is related to the use of automated tools such as Mace4 [11] for counterexample search in algebra, though we use Z3 for constraint solving and Lean 4 for independent formal verification.

Finite model theory and machine-checked proofs. Independence results via finite counterexamples are standard in finite model theory [9]. Each Cayley table is frozen in the Lean source and verified by the kernel, eliminating transcription errors. The sizes involved ($N = 5$ through $N = 10$) are small enough for `decide`/`native_decide` but large enough that hand-verification of all conditions would be error-prone. The universal algebraic theorems (no-associativity, decomposition invariance, ICP equivalence) are proved by pure equational reasoning without `decide`, and hold for all finite sizes.

9 Discussion

The classifier dichotomy is organizational. D organizes elements into coherent categories: classifiers that test and non-classifiers that transform. The $H \not\Rightarrow D$ result shows that an algebra can internalize composition without this organization. D contributes semantic structure rather than compositional behavior.

Why magmas? Magmas are not a limitation but a natural choice, for two independent reasons. First, the K-infinity theorem (Theorem 3.7) shows that nontrivial combinatory algebras with **s** and **k** must be infinite: the **k** combinator is a non-surjective injection on any finite set with ≥ 2 elements. Finite self-description requires a different combinator basis, one without a K-like projector, and our extensional 2-pointed magmas provide exactly this. Second, Theorem 3.6 shows that associativity forces any classifier to have a constant row, collapsing it to an absorber. Combined with the no-right-identity result (Theorem 3.5), this independently rules out the entire classical algebraic hierarchy: no semigroup, monoid, group, or ring can support the classifier dichotomy alongside a retraction pair. Self-description lives outside both the PCA/TCA framework (by cardinality) and the classical algebraic hierarchy (by associativity). This connects to the broader observation that reflective systems are inherently non-associative: λ-calculus application is non-associative ($(f\,g)\,x \neq f\,(g\,x)$ in general). The no-associativity theorem shows this is algebraically forced, not a design accident.

The role of extensionality. Extensionality (distinct elements have distinct rows) is load-bearing throughout. Without it, the no-associativity proof fails: Step 4 derives $\tau = v$ from identical rows. Without it, any constant-row element trivially satisfies both classifier and non-classifier conditions, making the three-category decomposition degenerate. Extensionality is not merely a convenience but the condition that makes element-level classification meaningful. An element's identity *is* its row in the Cayley table; extensionality formalizes this.

Why these three properties? R, D, and H are three algebraically simple and independently motivated properties in the landscape carved out by the obstructions. K-infinity and no-associativity rule out all classical settings; what remains are finite non-associative extensional magmas with absorbers. In this setting, the three basic structural questions are: can the algebra encode its own elements (R)? Can elements be cleanly partitioned by behavioral type (D)? Does the left-regular representation contain a non-trivial factorization (H)? The independence theorem confirms these are genuinely three questions, not one. Each property also connects to a standard algebraic or categorical concept. Capability R is a section-retraction pair [7]: two core elements s, r satisfying $r \circ s = \mathrm{id}$ on core. Capability D is a dichotomic purity condition on the left-regular representation: every core element's action on core is either fully absorber-valued or fully non-absorber-valued, with no partial classification. This isolates the Boolean aspect of a subobject classifier [8]: classification takes values in a two-element set with no intermediate truth values. It does not capture the universal aspect. In concrete witnesses, classifiers classify a fixed partition of core (2 of $2^{|\mathrm{core}|}$ subsets), not arbitrary subobjects, and the left-multiplication maps are not closed under composition, so the categorical framework in which to state a universal property does not arise. D is a separation condition on the morphism space, not a universal property. Capability H asks for a non-trivial factorization in the left-regular representation $\{L_a\}$. The no-associativity obstruction (Theorem 3.6) predicts that associative categorical frameworks will not apply to either connection; this is confirmed by the composition-closure failure noted above. Whether D corresponds to a known concept in non-associative algebra, or constitutes a genuinely new structural property, remains open. Whether H is equivalent to a standard categorical universal property also remains open.

Why the independence holds. The independence of D and H reflects incompatible row constraints under finiteness. A classifier's row is fully committed to absorber values on core: every core input maps to z_1 or z_2. An ICP witness a requires $a \cdot x = c \cdot (b \cdot x)$ with at least two distinct values on core, so its row must be non-trivially structured. These demands are incompatible for any single element, so different elements must carry these roles. The independence holds because finiteness limits the total supply of elements and constrains which row patterns can coexist under extensionality. The $H \not\Rightarrow D$ direction has the same character: an ICP triple (a, b, c) constrains three rows to satisfy a factorization equation, but this imposes no absorber-output condition on any fourth element.

The infinite boundary. In infinite carriers where the set of left-multiplication maps is unconstrained, such as the full function space over a countable set, the row-scarcity constraint dissolves. The carrier is large enough to accommodate elements with arbitrary row patterns, and the tension between classifier rows and factorization rows need not arise. Whether the independence survives in specific infinite algebraic settings is open. This is an observation about resource limitations, not a theorem about Turing completeness: the precise conditions under which the independence transfers require defining the relevant algebraic properties for infinite carriers, which is beyond this paper's scope.

R as a separate design choice. The independence of R from D and H plausibly extends beyond finite magmas. A faithful internal encoding, a section-retraction pair on the carrier, is a separate design choice from classification or composition. Mogensen's self-interpreter for the λ-calculus [5] requires an explicit quoting convention; without it, the calculus is computationally universal but lacks internal self-representation. This suggests that the R independence is not an artifact of finiteness but reflects a genuine structural distinction between having computational capabilities and having a faithful encoding of one's own elements as data.

Self-simulation. R, D, and H can be viewed as algebraically separable ingredients in one natural architecture for self-simulation (computing an algebra's own operation table from within): R provides encoding, D provides classification, and H provides internal composition. Formalizing non-trivial self-simulation in finite total magmas is an open problem. Non-trivial self-simulation in the PCA [13] or Turing category [17] sense requires a fixed evaluation mechanism and universality over an open-ended domain [6]. Finite total magmas lack both: the evaluator is the magma operation itself, the domain is the n^2 table entries (closed and finite), and the undecidability that prevents triviality in infinite settings is absent. This is the same structural feature that produces the K-infinity and no-associativity obstructions. A companion artifact demonstrates a reflective tower built on a 16-element algebra satisfying R, D, H with additional computational axioms (Branch, Y-combinator); the full computational system is described elsewhere.

Scope and limitations. Whether the three-capability separation appears in other settings (λ-calculus, typed systems, infinite algebras) is the primary open question. Theorem 3.6 constrains the target: any such translation must work without associativity or identity. A categorical characterization of H as a universal property (stronger than the analogy above) is also open. Two further directions connect to the frameworks discussed in Section 8: (1) the *extraction theorem*: do the emergent behavioral roles (absorbers, classifiers, non-classifiers) formally yield a Turing category when used as objects, with role-respecting elements as morphisms? (2) the *realizability connection*: does the classifier dichotomy, viewed as a property of an underlying PCA, predict specific logical principles in the associated realizability topos?

The cost of legibility. Additional organizational axioms push the minimum coexistence size to $N = 12$ (SAT-verified) and produce interpretable role structure; these are explored in a companion paper.

Realization freedom. Each capability admits multiple axiom forms; the independence result holds for all variants.

10 Conclusion

Two obstructions, K-infinity and no-associativity, rule out both combinatory algebras and the classical algebraic hierarchy as settings for finite algebras combining encoding with classification. In the remaining landscape of finite extensional 2-pointed magmas, we have identified three natural algebraic properties, self-representation (R), the classifier dichotomy (D), and the Internal Composition Property (H), and proved they are pairwise independent. The six non-implications are machine-checked in Lean 4. The optimal coexistence witness has $N = 5$ (Theorem 7.2). The three-category decomposition induced by D is a functorial invariant (Theorem 4.2), and the ICP is logically equivalent to Compose+Inert (Theorem 5.1). All results are formalized in Lean 4 (80 theorems across 10 files, zero `sorry`).

Open problems.

(1) *Cross-formalism universality*: Does the three-capability separation appear in the λ-calculus, typed settings, or infinite algebras? The no-associativity obstruction (Theorem 3.6) constrains the target setting.
(2) *Categorical characterization of H*: Is the ICP equivalent to a standard universal property (e.g., existence of a partial internal hom)?
(3) *Categorical characterization of D*: The classifier dichotomy captures the Boolean/decidability aspect of a subobject classifier (purity of classification, no intermediate truth values) but not the universality aspect (unique characteristic morphism for every subobject). The left-multiplication maps lack composition closure, precluding standard categorical formulations. Does D correspond to a known concept in non-associative categorical algebra, or does it constitute a new structural property?
(4) *Internal self-simulation*: Does there exist a finite extensional 2-pointed magma satisfying R, D, and H (with the Branch and fixed-point enrichments) that supports non-trivial self-simulation: an internal evaluator that computes its own Cayley table, or a non-trivial fragment thereof, entirely from within the magma? Non-trivial self-simulation in the PCA or Turing category sense requires a fixed evaluation mechanism and universality over an open-ended domain, both absent in the finite total setting. Whether a meaningful intermediate definition exists is open.

Artifact. All Lean files, counterexample tables, SAT reproduction scripts, and the 16-element reflective tower artifact are available at https://github.com/stefanopalmieri/finite-magma-independence.

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A Cayley Tables

All tables are extracted from the Lean source files and verified by `lake build`. Element i's row is row i; entry (i, j) is $i \cdot j$.

$N = 4$*: Minimal R+D witness ($s = r$).* Roles: $0 = z_1$, $1 = z_2$, $2 = \tau$, $3 = s = r$. Lean: `CatKripkeWallMinimal.lean`, `kripke4`.

·	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	0	1	0	1
3	0	0	2	3

$N = 5$*: Minimal R+D witness ($s \neq r$).* Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. Lean: `CatKripkeWallMinimal.lean`, `kripke5`.

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	1	0	3	4	2
3	0	2	4	2	3
4	0	1	1	0	0

$N = 5$*: R+D+H Optimal Coexistence ($s = r$, self-simulates).* Roles: $0 = z_1, 1 = z_2, 2 = s = r$ (identity on core, ICP b), $3 = \tau$ (classifier, ICP a, self-sim term), $4 = \tau_2$ (classifier, ICP c). Lean: `Witness5.lean`, `SelfSim5.lean`.

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	2	2	3	4
3	0	0	0	1	0
4	0	1	0	1	0

$N = 6$*: R+D+H Coexistence (role overlap).* Roles: $0 = z_1$, $1 = z_2$, $2 = s$ (also ICP a), $3 = r$ (also ICP b), $4 = \tau$, $5 =$ ICP c. Lean: `Witness6.lean`.

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	3	3	4	2	5	3
3	0	1	3	5	2	4
4	0	0	1	0	1	1
5	2	2	5	4	3	2

$N = 8$*: $R \not\Rightarrow D$ Counterexample.* Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 =$ classifier, $5, 7 =$ mixed (violate dichotomy). Lean: `Countermodel.lean`.

·	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1
2	3	3	7	3	4	6	5	2
3	0	1	7	3	4	6	5	2
4	0	0	0	0	0	0	1	0
5	6	2	6	2	1	1	1	1
6	0	0	5	2	2	2	2	2
7	2	2	2	1	2	2	6	3

$N = 6$*: $R \not\Rightarrow H$ Counterexample (structural).* Roles: $0 = z_1$, $1 = z_2$, $2 = s = r$ (involution on core). Retraction pair holds; no triple satisfies ICP despite 4 core elements. Lean: `E2PM.lean` (`sNoH6_e2pm`).

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	0	3	3	2	5	4
3	2	4	5	5	1	4
4	5	3	0	0	3	2
5	4	2	2	2	2	2

$N = 10$*: $D \not\Rightarrow H$ Counterexample.* Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. No triple satisfies ICP. Lean: `Countermodels10.lean` (`dNotH`), `ICP.lean` (`dNotH_no_icp`).

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	3	2	3	4	5	6	7	9	8
3	0	1	2	3	4	5	6	7	9	8
4	0	0	1	1	1	1	1	1	1	1
5	1	0	0	0	0	0	0	0	0	0
6	2	3	9	9	9	9	9	9	9	8
7	3	2	9	9	9	9	9	9	9	8
8	1	0	1	0	1	1	1	1	0	0
9	0	1	0	1	0	1	1	0	1	1

$N = 10$*: $H \not\Rightarrow D$ Counterexample.* Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. ICP holds ($a = 8, b = 6, c = 7$). Element 5 violates dichotomy. Lean: `Countermodels10.lean` (`hNotD`).

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	1	3	4	9	6	8	5	7	2
3	0	1	9	2	3	7	5	8	6	4
4	0	0	1	1	1	1	1	1	0	0
5	0	0	2	0	0	0	0	0	3	1
6	2	2	2	8	3	9	4	7	9	7
7	8	3	2	8	3	9	4	7	3	1
8	9	2	2	3	8	1	3	7	1	7
9	2	2	2	2	4	7	6	7	2	0

$N = 5$*: $D \not\Rightarrow R$ Counterexample.* No retraction pair exists. Element 2 is a classifier: row on core is $(1, 0, 0) \subseteq \{z_1, z_2\}$. Elements 3 and 4 are non-classifiers: all core outputs ≥ 2. Dichotomy holds; no pair of core elements forms a retraction pair. Lean: `E2PM.lean` (`dNoS_e2pm`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	0	1	0	0
3	4	2	4	2	4
4	2	2	3	3	3

$N = 6$*: $H \not\Rightarrow R$ Counterexample.* No retraction pair exists. ICP holds: $a = 2, b = 4, c = 5$ with b core-preserving and $2 \cdot x = 5 \cdot (4 \cdot x)$ on core. Lean: `E2PM.lean` (`hNoS_e2pm`).

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	3	3	2	2	2	4
3	0	2	2	2	4	0
4	4	3	2	2	4	5
5	0	0	2	2	2	4

$N = 10$*: R+D+H Coexistence Witness (all roles distinct).* R+D+H roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$, $8 = \eta$ (ICP a), $6 = g$ (ICP b, core-preserving), $7 = \rho$ (ICP c). Enrichment roles (not used in this paper's theorems): $5 = f$, $7 = \rho$ (Branch dispatch), $9 = Y$ (fixed point). Lean: `Witness10.lean`.

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	3	4	3	7	5	9	6	8	2
3	0	1	9	3	2	5	7	4	8	6
4	0	0	1	1	1	0	0	0	1	1
5	2	2	7	2	8	9	4	3	4	2
6	0	0	6	4	8	7	3	3	4	9
7	2	2	6	4	8	9	4	3	4	9
8	2	2	4	8	4	3	4	4	8	9
9	3	4	7	3	9	2	2	9	2	3

B Proof Inventory

All Lean files compile with `lake build`, zero `sorry`. Proof styles: *Algebraic* = pure equational reasoning (no `decide`); *decide* = verified by kernel computation ($N \leq 8$); *native_decide* = verified by compiled native code ($N = 10$).

File	Thms	Style	Content
CatKripkeWallMinimal	17	Algebraic	Decomposition, bounds, wall
NoCommutativity	3	Algebraic	Asymmetry
Functoriality	4	Algebraic	Invariance under iso
ICP	20	Alg.+decide	ICP $\Leftrightarrow$ Compose+Inert
Countermodel	5	decide	$R \not\Rightarrow D$ ($N = 8$)
Countermodels10	9	native_dec.	$D \not\Rightarrow H$, $H \not\Rightarrow D$
E2PM	10	decide	$D \not\Rightarrow R$, $H \not\Rightarrow R$, $R \not\Rightarrow H$
Witness10	6	native_dec.	R+D+H at $N = 10$
Witness5	3	decide	R+D+H at $N = 5$, no ICP at $N = 4$
Witness6	3	decide	R+D+H at $N = 6$
Total	**80**		